

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

3rd April 2021

Subject CS1A – Actuarial Statistics (Paper A)

Time allowed: 3 Hours 30 Minutes (9:30 – 13:00 Hours)

Total Marks: 100

- Q. 1) i)** A company hires candidates from 4 institutes. It selects 20% of candidates from Institute A and B each. Balance candidates are selected from Institute C and D with equal share. Probability that selected candidate will join the company is 99% for Institute A, 98% for Institute B, 97% for Institute C and 96% for Institute D.

Calculate the probability in each of the following cases

- a Candidate who fails to join the company is from Institute A
 - a Candidate who fails to join the company is from Institute B
 - a Candidate who fails to join the company is from Institute C
 - a Candidate who fails to join the company is from Institute D. (3)
- ii)** Identify the incorrect statement/s, if any and correct them
- a) Under squared error loss, the mean of the posterior distribution minimises the expected loss function
- b) Under absolute error loss, the mean of the posterior distribution minimises the expected loss function
- c) Classical approach treats unknown parameter as a random variable in statistical estimation problems
- d) In classical statistics, unknown parameter is a fixed quantity and probability can be assigned to it (2)
- iii)** Calculate the Bayesian estimate of W under squared error loss when posterior distribution of W follows $\text{Gamma}(48, 4)$. (1)
- iv)** A student has correctly identified posterior distribution to be $\text{Ga}(a, b)$ and correctly calculated Bayesian estimate under squared error to be 'S'. Then, Bayesian estimate under Absolute error loss and under All-or-nothing loss must be
- a. Greater than 'S'
 - b. Lower than 'S'
 - c. equal to 'S'
 - d. none of the above (1)
- [7]

- Q. 2) i)** What is the main purpose of performing Factor analysis? Comment on the original variables and newly identified Principal components. (3)
- ii)** A student came up with 5 by 5 variance-covariance matrix of the Principal Components (PC1, PC2, PC3, PC4, PC5) with these 5 diagonal entries: 0.456, 0.137, 0.080, 0.0165 and 0.012 respectively. Identify the percentage of the total variance explained by each Principal Component. (4)
- iii)** What can be concluded based on the results of part (ii). (2)
- [9]

- Q. 3) i)** Calculate the probability that the sample variance of a sample of 5 values from a normal distribution will be more than 3 times the sample variance of a sample of 17 values from an independent normal distribution with the same variance.

You are given that

$F_{4,16}$ at 10% is 2.333

$F_{4,16}$ at 5% is 3.007 and

$F_{4,16}$ at 2.5% is 3.729. (3)

- ii)** In case of Kendall's correlation, Identify the number of observations in the assignment, given that there are no ties and $N_c = 87$ and $N_d = 123$?

Note – N_c and N_d denote number of concordant and discordant pairs respectively.

- a. 18
- b. 19
- c. 20
- d. 21
- e. 22

(2)

- iii)** State the properties that can lead to data being classified as 'big' data. (2)

[7]

- Q. 4) i)** If coefficient of variation is defined as

Coefficient of variation = Standard variation / mean

What will be the effect of an increase in variance on Coefficient of variation in case of following distributions?

- a) Chi Square Distribution
- b) Poisson Distribution
- c) Exponential Distribution

(6)

- ii)** Using inverse transform method, simulate 2 values from $\text{Exp}(0.5)$ distribution using random value 0.769 and 0.004 from $U(0,1)$ distribution. (2)

[8]

- Q. 5) i)** For the following joint distribution between X and Y, Calculate $E[Y|X=2]$

		Y		
		1	2	3
X	1	0.1	0.1	0.3
	2	0.3	0.2	0

(2)

- ii)** If X and Y are independent standard normal variables then calculate the mean, variance and the distribution of $5X - 4Y$. (2)

- iii)** If a random variable X follows Gamma (10,0.1) then $P(X>50)$ is equivalent to finding the probability using the certain chi square distribution. Identify the equivalent

probability calculation formula involving chi square distribution and calculate the required probability using chi square tabulated value.

You are given that chi square tabulated value is 0.0318.

(3)

iv) Calculate the required probability as mentioned in Part (iii) above using normal approximation.

(3)

v) Comment on the results obtained in part (iii) and (iv).

(2)

[12]

Q. 6) i) Claim amounts $x_1, x_2, \dots, x_n$ are observed over n years. Claim amounts over 6 years equal 3600.

If, Total claim amounts per annum follow $N(b, 100^2)$ and Prior believes are that b is $N(700, 70^2)$.

Identify mean, variance of posterior distribution of β

(2)

ii) Using the information from earlier part, calculate the posterior probability that b is greater than 600 using normal-normal model.

(2)

iii) Comment on the possible value of credibility factor Z and true mean based on the results in part (i).

How would the value of Z change with change in standard deviation of prior and likelihood?

(3)

[7]

Q. 7) An insurance company has a sample of 500 motor insurance policies with claims. Say X denotes the number of policies where the annual claim amount exceeded INR 10,000 and X follows Binomial distribution with parameters (5000, p).

i) Assuming annual total claim amounts per policy are independent and identically distributed, estimate the unknown proportion " p " of claims higher than 10,000 per year by deriving its maximum likelihood.

(4)

ii) If there is a prior knowledge about p follows Uniform distribution with parameters (0, 1) then derive the density of the Bayesian posterior distribution of p in terms of n and X . Also state the name of the posterior distribution that p follows along with its parameters.

(4)

If 200 out of 500 policies with claims from the sample have a claim above INR 10,000 then

iii) Estimate p using the MLE in (i).

(1)

iv) Estimate p using the Bayesian estimator under quadratic loss, based on the posterior distribution derived in part (ii).

(2)

v) Comment on the difference between the values of p estimated in (iii) and (iv)

(1)

vi) From part (iv), express the Bayesian estimator in the form of credibility factor and determine the value of credibility factor.

(3)

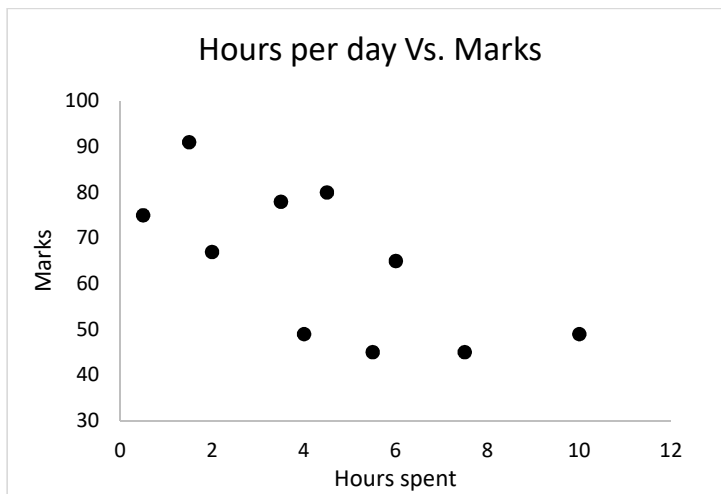
[15]

Q. 8) The below data gives the marks scored and time spent on social media in hours per day:

Hours x	6	5.5	0.5	3.5	7.5	4.5	4	1.5	2	10
Marks y	65	45	75	78	45	80	49	91	67	49

For this data: $\sum x = 45$; $\sum x^2 = 277.5$; $\sum y = 644$; $\sum y^2 = 43,956$; $\sum xy = 2,602$

Scatterplot for the above data is provided below:



- i) Based on the above scatterplot, comment on the association between marks scored and hours spent per day on social media. (1)
- ii) Calculate the correlation co-efficient between the two variables and comment on the same. (4)
- iii) Investigate the hypothesis that there is a negative correlation using Fisher's transformation of the correlation co-efficient. You should clearly state the hypothesis of your test and any assumption made for the test to be valid. (7)
- iv) Fit a linear regression model to this data with marks being the response variable and hours spent as the explanatory variable. (3)
- v) Calculate the coefficient of determination for this model and interpret the same. (2)
- vi) Calculate the expected change in marks for every 1 additional hour spent on social media basis the model fit in (iv). (1)

[18]

Q. 9) Let $X_1, X_2, X_3, \dots, X_n$ be a random sample of claim amounts which follow gamma distribution with a parameter $\alpha = 5$ and unknown parameter λ .

- i) Specify the distribution of $\sum_{i=1}^n X_i$ with parameters. Hence also state the distribution of $2n\lambda\bar{X}$. (3)
- ii) A random sample of 10 such claims has a mean of 100. Pick the 95% confidence interval for λ .

Option A: (0.03897, 0.06220)

Option B: (0.03711, 0.06480)

Option C: (0.01618, 0.03570)

Option D: (0.01738, 0.03375)

(3)

[6]

Q. 10) An insurance company models claim numbers on health insurance portfolio using a Poisson distribution and the mean depends on age and gender of policyholder.

i) Specify the link function for fitting a generalized linear model for the mean of the Poisson distribution

(1)

ii) State the linear predictor used for modelling the age and gender dependence as:

a) Age + Gender

(2)

b) Age + Gender + Age . Gender

(2)

[5]

Q. 11) In a survey by Industry, 20 organizations from each industry were questioned on their attrition rate and a summary was prepared as below:

Industry	Manufacturing	IT / ITES	Consulting
Average resignation %	27%	36%	30%
Sample standard deviation	5%	10%	8%

Perform a one-way analysis of variance test to test the hypothesis that the Industry has no impact on resignation rate.

[6]
